EXP NO:7 Random Sampling and Sampling Distribution

AIM:
To understand random sampling and the sampling distribution of the sample mean.

PROGRAM:

python

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Creating a population of 10000 values (right-skewed data)
np.random.seed(42)
population = pd.Series(np.random.exponential(scale=50, size=10000))

print("Population Mean:", population.mean())
print("Population Standard Deviation:", population.std())

# Histogram of the population
plt.hist(population, bins=40, color='skyblue', edgecolor='black')
plt.title("Population Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()

# Simple random sample of size 50
sample = population.sample(n=50, random_state=1)
print("Sample Mean:", sample.mean())
print("Sample Standard Deviation:", sample.std())

# Sampling distribution: take 1000 random samples of size 50
sample_means = []

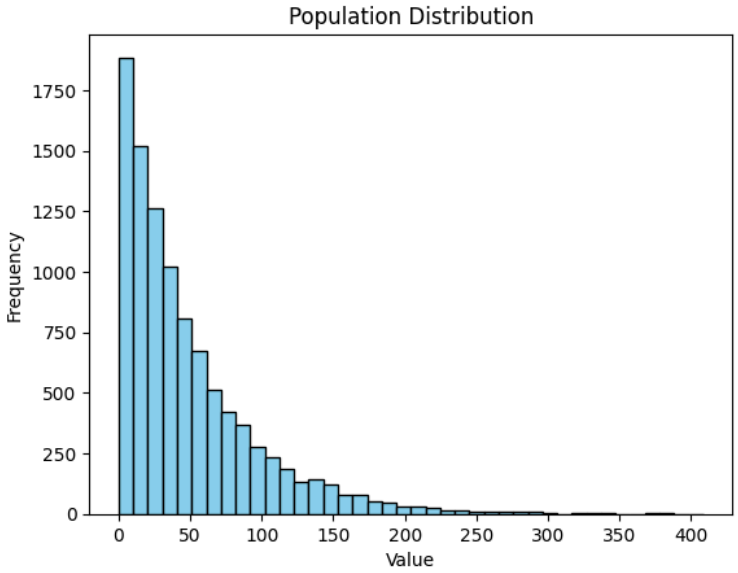
for i in range(1000):
    s = population.sample(n=50)
    sample_means.append(s.mean())

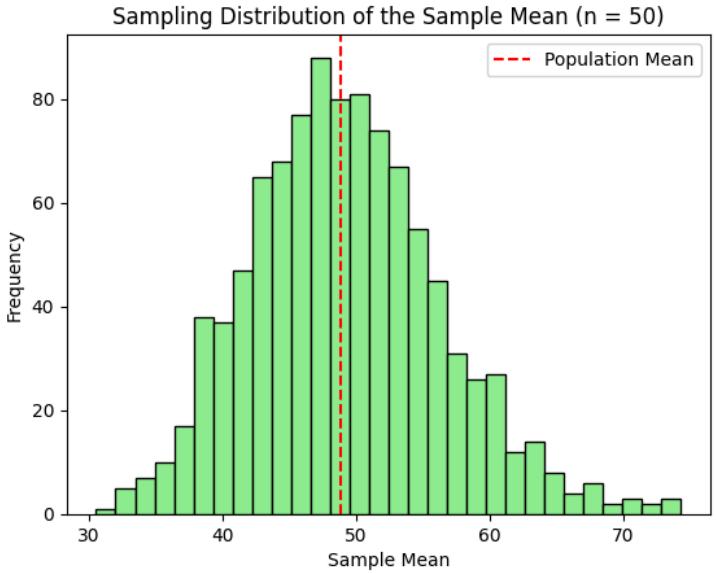
sample_means = np.array(sample_means)

print("Mean of Sample Means:", sample_means.mean())
print("Standard Error (std of sample means):", sample_means.std())
print("Theoretical Standard Error:", population.std() / np.sqrt(50))

# Histogram of the sampling distribution
plt.hist(sample_means, bins=30, color='lightgreen', edgecolor='black')
plt.axvline(population.mean(), color='red', linestyle='--', label='Population Mean')
plt.title("Sampling Distribution of the Sample Mean (n = 50)")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```

OUTPUT:





RESULT:

Thus the program to understand random sampling and the sampling distribution of the sample mean was executed successfully.